Troposkein VAWT experimental dataset review

Prepared for discussion with Mridul | 24 October 2026

Objective: assemble experimental Cp versus tip-speed-ratio data and investigate whether geometry and normalization explain the offset observed for the Sandia 17-m turbine.

Recommendation

Begin with the Sandia 2-m report, the DeepWind numerical tables and Battisti 2018 Figure 10. Pursue the Sandia 5-m, 34-m and NASA reports next. Treat thesis and compilation leads as retrieval tasks until their actual curves are checked.

What this review verifies

This is a source-screening review, not a completed curve extraction audit. Primary report records, selected full-text passages, Battisti 2018 Figure 10 and DeepWind Tables 1-4 were checked. No user CSVs were supplied for this review, so no extracted points or filename-to-curve mappings have been validated.

Evidence status key

Tables / figure verified: The stated extraction target was located in full text. Geometry and curve metadata may still need checking.

Configurations verified: Experimental setup details were checked, but a complete figure inventory is pending.

Report record verified: An institutional record confirms the publication and broad scope; exact curves remain unchecked.

Lead only / bibliographic lead: A retrieval candidate, not an approved dataset.

Supporting: Methods, geometry or duplicate checks; not automatically an additional independent dataset.

Scope decision

Keep experimental curved-blade troposkein-family rotors, recording exact versus approximated blade shape. Exclude H-rotor curves from the troposkein collection itself. Keep tilted cases separately identifiable. Retain one canonical copy of each measurement series.

1. Verified extraction targets

Battisti2016 | DeepWind demonstrator

Data in Brief 8, 1120-1126; DOI 10.1016/j.dib.2016.07.029

Evidence: Tables verified

Content: Three-bladed troposkien rotor; numerical torque coefficient versus equatorial TSR.

Extraction target: Tables 1-2: upright at 200/300 rpm. Tables 3-4: 15-degree tilt at 200/300 rpm.

Properties: Reported rotational chord Re: 138000 at 200 rpm; 205000 at 300 rpm. These are not Re_D.

Audit action: Derive Cp = TSR * C_Q. Companion Energy paper shares measurements. Preserve negative torque points in original data; any analysis exclusion must be recorded.

Open source / repository record

Battisti2018 | Troposkien and H-shaped comparison

Renewable Energy 125, 425-444; DOI 10.1016/j.renene.2018.02.098

Evidence: Figure verified

Content: Direct experimental comparison with the same swept area and blade section.

Extraction target: Figure 10: Cp versus equatorial TSR. Keep troposkien curves; H-shaped curves are comparison-only.

Properties: Uses blade-area/swept-area solidity. Reported solidity: H-shaped 0.25; troposkien 0.38.

Audit action: Same facility does not isolate radius effects. Check shared DeepWind hardware and overlapping conditions before merging campaigns.

Open source / repository record

DeepWind conversion: Cp = TSR_eq x C_Q,aero follows from P = Q x omega and the paper's matching area and radius definitions. This produces four table-based curves, not four separate experimental campaigns. The 2016 Energy companion should supply supporting methods without duplicating these curves.

2. Sandia experiments

Blackwell1976 | Sandia 2-m

SAND76-0130; DOI 10.2172/7269797

Evidence: Experimental configurations verified

Content: NACA 0012 rotor tested with two and three blades and different chords.

Extraction target: Full figure-by-figure extraction list still required.

Properties: Straight-line/circular-arc blade geometry approximates a troposkein.

Audit action: Record approximation rather than labeling it an exact mathematical troposkein. Verify chord and conditions per curve.

Open source / repository record

Sheldahl1980 | Sandia 5-m

SAND80-0179; DOI 10.2172/5303846

Evidence: Report record verified

Content: Sheldahl, Klimas and Feltz: aerodynamic performance with extruded aluminum NACA 0015 blades.

Extraction target: Exact Cp-TSR figures need full-text audit.

Properties: Blade counts, chords and Reynolds definitions remain to be extracted.

Audit action: Check whether earlier 5-m measurements are repeated.

Open source / repository record

Ashwill1992 | Sandia 34-m

SAND91-2228

Evidence: Report record verified

Content: Performance across wind conditions and rotation rates; analytical comparisons also included.

Extraction target: Exact Cp-TSR figures not yet audited.

Properties: Variable-speed field test; step-tapered geometry flagged in supplied lead, requiring full-report verification.

Audit action: Do not assign one chord or import analytical curves. Check overlap with earlier 34-m reports.

Open source / repository record

3. Additional experiments

Muraca1976 | NASA 14-ft rotor

NASA TM X-72663

Evidence: Report record verified

Content: Muraca and Guillotte: wind tunnel measurements of wind speed, shaft torque and rotation rate.

Extraction target: Generalized performance against TSR; exact Cp figure numbers need audit.

Properties: Nominal diameter 14 ft = 4.2672 m.

Audit action: Measured and analytical results coexist. Retain measured performance only; verify blade geometry and normalization in full report.

Open source / repository record

Sheldahl1977 | Sandia 5-m

SAND77-1063; DOI 10.2172/5265052

Evidence: Report record verified

Content: Sheldahl and Blackwell: free-air performance tests of a 5-metre-diameter Darrieus turbine.

Extraction target: Exact Cp-TSR figures need full-text audit.

Properties: Earlier 5-m experiment than SAND80-0179.

Audit action: Compare blade construction, geometry and campaigns with 1980 report; distinct publication alone does not establish independent curves.

Open source / repository record

WorstellAddendum | Sandia 17-m, three blades

Three-bladed configuration: an addendum; OSTI 5477166

Evidence: Report record verified

Content: Potential blade-count extension to the existing 17-m collection.

Extraction target: No figures verified; OSTI record indicates no digital full text.

Properties: Three-bladed configuration is explicit in report title.

Audit action: Compare with existing 17-m files before adding. Date and exact report number still need verification.

Open source / repository record

4. Supporting publications and leads

Battisti2016Energy | DeepWind demonstrator

Energy 111, 484-497; DOI 10.1016/j.energy.2016.05.080

Evidence: Companion publication verified

Content: Wind tunnel testing in upright and tilted operating conditions.

Extraction target: Use for methods and geometry; exact additional nonduplicate curves need audit.

Properties: Same experimental campaign as Data in Brief companion.

Audit action: Do not count both publications as independent experiments.

Open source / repository record

Read1986 | Curved-bladed rotor thesis

Kingston Polytechnic thesis

Evidence: Lead only

Content: A theoretical and experimental study of the aerodynamics of the curved-bladed Darrieus vertical axis wind turbine.

Extraction target: No figures verified.

Properties: Two blades and chord Re 28000 are claims in the supplied lead, not verified inputs.

Audit action: Do not populate numeric metadata from the lead. Retrieve thesis first.

Open source / repository record

SouthRangi | NRC Canada rotor tests

Exact report to be identified

Evidence: Lead only

Content: Earlier Canadian experiments referenced by later literature.

Extraction target: No primary figures verified.

Properties: Geometry, blade counts and conditions unverified.

Audit action: Obtain original report; avoid digitizing secondary reproductions as independent data.

Primary source link not yet established.

VeersCompilation | Sandia selected papers

SAND90-1615

Evidence: Bibliographic lead

Content: Compiled papers potentially covering 34-m design, initial performance and test status.

Extraction target: Individual chapters and performance figures not verified.

Properties: Do not treat the volume as a new rotor or independent campaign.

Audit action: Check each chapter against Ashwill1992 before selecting unique curves.

Open source / repository record

5. Geometry and normalization audit

Solidity definitions must remain distinct

The existing pipeline uses sigma_circ = N c / (2 pi R). Battisti 2018 uses sigma_area = N c L / A, where L is blade length and A is projected swept area. For variable chord, the corresponding blade-area definition is sigma_area = N integral[c(s) ds] / A. These are distinct conventions, not interchangeable entries in a single unlabeled column. [Battisti2018, equations 4-7]

For an upright axisymmetric curved rotor, A = integral[2 r(z) dz]. Do not automatically substitute maximum diameter times height. If an area convention changes, Cp_new = Cp_old x A_old / A_new, provided power, density and wind speed are unchanged. This is a normalization conversion, not a fit to other turbines.

Radius and TSR

With consistent wind speed and angular speed, TSR_mean = (r_mean / R_max) TSR_max. The averaging rule for r_mean must be specified: height-, blade-length- and other weighted averages differ. A change to TSR alone does not change measured Cp. Battisti 2018 discusses equatorial TSR and the continuous variation of local blade speed with height.

Reynolds number

Use Re_D = U D / nu = rho U D / mu, with D = 2 R_max when that is the chosen diameter. nu is kinematic viscosity (m2/s); mu is dynamic viscosity (Pa s). Both the fluid conditions and the velocity/length basis must be documented.

DeepWind reports Re_rot,c = omega R_max c / nu. With TSR = omega R_max / U, the conversion is Re_D = Re_rot,c x D / (TSR x c). It is valid only for those definitions and known geometry. At fixed rpm with changing wind speed, Re_rot,c can be constant while Re_D varies along a curve. Do not put one constant Re_D into that CSV unless justified.

The 17-m question remains open

The exact multiplier and the quantity corrected were not provided. No correction was applied in this package. Compare raw versus normalized Cp, swept area, radius, solidity convention, wind-speed reference, mechanical/electrical power, parasitic losses and Reynolds basis before attributing the offset to a universal shape factor. Use other rotors to test a physically derived factor, rather than fitting each rotor to agree.

Required from the existing 17-m analysis: report and figure; original curve; multiplier; corrected axis or parameter; and the geometric derivation, if any.

6. Metadata and assumptions workflow

Files in this package

troposkein_sources.csv: one record per candidate publication, alphabetized by Source_ID. It includes verification status, extraction target, audit notes and source link. It is a tracking table, not performance data.

metadata_troposkein.csv: header-only curve metadata template. Add one row per accepted experimental curve after verification. No artificial curves or unverified geometry values have been inserted. Compatibility with the latest Colab importer has not been tested.

Adding future curves

Assign a permanent Curve_ID (C001, C002, ... within this separate collection) once accepted. Sort by source, numeric figure and panel without renumbering existing IDs. Keep a concise filename such as SourceYear (Fig 10a) C001.csv. Conditions belong in metadata. Check the complete Windows path length, not just the filename.

Use TSR and Cp as numeric performance columns. Add N and cToR to the performance CSV only if the current importer requires them. A metadata file does not automatically join itself to the notebook; use CSV_filename or Curve_ID as an explicit join key. Add conditions that vary point-by-point, such as Re_D during a fixed-rpm wind-speed sweep, to the performance CSV rather than forcing a constant metadata value.

Minimal documentation rules

Unknown values stay blank, not zero. Preserve reported Reynolds and solidity definitions before deriving normalized values.

For constant chord, cToR means c / R_max unless an explicit alternate basis is recorded. For variable chord, leave it blank until an approved effective-chord definition exists.

Record viscosity assumptions with medium, temperature, pressure when relevant, mu and/or nu, density if required, source and rationale. No numerical viscosity has been assumed in this package.

Keep blade pitch alpha_p separate from whole-rotor Tilt_deg. Preserve the paper's pitch sign convention; do not assume all authors use the same sign.

Write each assumption or change in Assumptions_changes: property; reported value; adopted value; reason; evidence/page; date. Include exclusions and duplicate decisions.

Keep original points intact. Negative Cp can represent net power consumption; do not call it a digitization error solely because it is negative. Any generating-only subset should be a documented analysis filter.

Cp above 0.593 is a review flag, not an automatic deletion rule. Check blockage, augmentation and reference-area definitions before judging validity.

Outstanding approvals / evidence

Confirm the precise 17-m correction; obtain full reports for unverified figures; resolve exact versus approximated troposkein scope; verify variable-chord geometry; and choose a common solidity definition before calculating the comparison parameter.